

# SOWMYA YALAVARTHI

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**Data Scientist** with 4+ years of experience in data engineering and advanced analytics, specializing in ETL pipelines, statistical/econometric modeling, and machine learning. Proficient in Python, R, SQL, and PySpark with expertise in Oracle, Snowflake, and Redshift. Skilled at anomaly detection, univariate/bivariate analysis, and NLP-driven predictive modeling, delivering insights that improve decision-making across energy, insurance, and healthcare.

## SKILLS

- **Programming:** Python, R, SQL, Shell Scripting, Git
- **Data & ML:** PySpark, Pandas, NumPy, Scikit-learn, XGBoost, LightGBM, NLP, Time Series, Neural Networks, SHAP, Econometric Modeling, Random Forest
- **Databases/ETL:** Oracle, Snowflake, Redshift, PostgreSQL, AWS S3, AWS Glue, Talend, Autosys
- **Visualization & Tools:** Power BI, Streamlit, Plotly, Matplotlib, Seaborn, MLflow, Agile

## EXPERIENCE

**Tern** **01/2025 – Present**  
Data Scientist New York

- Performed EDA on large global energy datasets, applying univariate and bivariate analysis with feature engineering to uncover demand shifts, anomalies, and climate-linked usage patterns for strategic planning.
- Designed KPIs and predictive models in Python and Power BI, quantifying sector-wise energy changes and enabling executives to make data-driven infrastructure investment decisions.
- Built reusable PySpark pipelines to standardize transformations, improving data consistency and model accuracy while reducing preparation time by 25%.

**Tata Consultancy Services** **06/2023 – 09/2023**  
Data Engineer Hyderabad, India

- Developed ETL pipelines using Python and SQL to centralize reporting in PostgreSQL, cutting report latency by 30% and supporting predictive analytics for a retail client.
- Conducted data profiling, validation, and schema refinement, ensuring data integrity for dashboards and downstream econometric and statistical modeling.
- Partnered with cross-functional teams to define KPIs and translate business rules into optimized SQL workflows, improving the accuracy of reporting and anomaly detection.

**Capgemini** **12/2019 – 05/2023**  
Data Engineer Hyderabad, India

- Migrated U.S. insurance data (The Hartford) from Oracle to Snowflake via PySpark and AWS S3, processing 5M+ records daily; tuned pipelines for a 10x performance gain.
- Converted PL/SQL packages to Snowflake SQL and implemented anomaly detection logic, reducing reporting errors by 30% and strengthening fraud detection analytics.
- Automated ETL workflows with Autosys and integrated CI/CD (Jenkins/uDeploy), ensuring reliable ingestion pipelines for statistical, econometric, and network modeling use cases.

## ACADEMIC PROJECTS

**Climate Aware Crop Yield Forecasting ([Github](#))** **05/2025**

- Built XGBoost and Random Forest models on 50K+ climate records ( $R^2 = 0.89$ ) and applied SHAP explainability to identify key drivers (rainfall, CO<sub>2</sub>, temperature), deploying a Streamlit app for scenario testing.

**Healthy Lifestyle Buddy – AI Chatbot ([Github](#))** **12/2024**

- Developed a mental health chatbot with Gemini API and Hugging Face NLP, integrated into Streamlit with sentiment analysis and journaling features, boosting engagement by 30% and earning Google Hackathon recognition.

## EDUCATION

**Pace University**, M.S in Data Science **09/2023 - 05/2025**  
**KL University**, Bachelors in Electronics & Communication Engineering **07/2015 - 05/2019**

## CERTIFICATIONS & ACHIEVEMENTS

- AWS Cloud Practitioner (CLF-C02), Google Hackathon (2025), Capgemini Customer Delight Award